

Final Project Proposal

Team Name: (deep) fake it till you make it

Srinidhi Ayalasomayajula (sa4564), Reed Evans (re2449), Gillian Simpson (gas2172)

Project Description

For our project, we are looking to extend Gillian's midterm paper which investigated privacy preservation tools for AI image and text generation. One of her findings was that these tools are largely proactive (intended to be deployed by developers) rather than defensive (for use by individuals), and that the defensive tools that do exist are largely intended for detection rather than prevention. Of these, there is a larger focus on preventing deepfakes and photorealistic images than copyright infringement mitigation for artists, which is still being litigated in courts.

To explore this topic further, we are looking to apply the anti-AI image generation tool Glaze to combat AI art video generation [1]. Specifically, we would like to explore if different video generation models respond differently to defensive perturbations. We would like to evaluate whether Glaze-based protections are consistently effective across multiple systems or maybe whether some models are able to bypass or ignore these perturbations. Our goal is to assess the robustness and transferability of defensive techniques in the video generation area.

Motivation

AI generated videos have become increasingly popular in recent years, including videos of fake actors, and AI-generated animation [2,3]. There are a number of tools that have been proposed to cloak real photos and videos from deepfakes [4,5], but few to prevent stylistic mimicry for art, and only for images [1]. Our goal is to apply a pre-existing tool for anti-AI style mimicry in images to videos. Users of this novel tool could apply it to their own animations and art videos to protect them from AI generation tools. Alongside further researching privacy preservation tools for AI imaging and text generation, we hope to empirically test our topic by experimenting with anti-AI image tools on the market.

Methodology

We will investigate what percent of frames must have Glaze applied in it in order to prevent artistic style copying via video. We will use some open-source animations online as well as create some simple ones to test our method against three text-to-video generation tools. We will evaluate its effectiveness through relation metrics of frames comparing the original and AI-generated videos trained on the Glazed video, similar to [4]. We will also apply some video similarity metrics specific to video and animation [6,7], and potentially surveys as well to compare against human metrics.

We will use a combination of open source animation clips and possible self created animations as the target styles we are looking to “protect”. Then, we are looking to apply Glaze to the video data by protecting all the frames. After the defensive measures are put in place, we would like to evaluate 3 text-to-video systems: Runway Gen-2 (commercial product), Pika Labs (stylization focused model), and ModelScope text-to-video (research based, open source). For each of the models, we provide prompts to replicate the style of the original video and generate outputs using the original videos and Glaze-protected videos. In the end, we will compare how well each model reproduces the original style under both conditions. Our evaluation metrics include a quantitative approach by analyzing things like frame level similarity, a style mimicry analysis that measures how closely the output videos match the original and a more general human evaluation that may include a user study to gather their opinions.

References

- [1] S. Shan, J. Cryan, E. Wenger, H. Zheng, R. Hanocka, and B. Y. Zhao, "Glaze: Protecting artists from style mimicry by Text-to-Image models," in 32nd USENIX Security Symposium (USENIX Security 23). Anaheim, CA: USENIX Association, Aug. 2023, pp. 2187–2204. [Online]. Available: <https://www.usenix.org/conference/usenixsecurity23/presentation/shan>
- [2] C. Murray, "Controversial AI 'Actress' Tilly Norwood Rebuffs Backlash In New Song," *Forbes*, Mar. 10, 2026. Available: <https://www.forbes.com/sites/conormurray/2026/03/10/ai-actress-tilly-norwood-responds-to-backlash-in-debut-song-ais-not-the-enemy/>
- [3] Szarowicz, A., Amiguet-Vercher, J., Forte, P., Briggs, J., Gelepithis, P., Remagnino, P. (2001). The Application of AI to Automatically Generated Animation. In: Stumptner, M., Corbett, D., Brooks, M. (eds) AI 2001: Advances in Artificial Intelligence. AI 2001. Lecture Notes in Computer Science(), vol 2256. Springer, Berlin, Heidelberg. https://doi.org/10.1007/3-540-45656-2_42.
- [4] E. Guo, P. Li, S. Yu and H. Wang, "Efficient Video Privacy Protection Against Malicious Face Recognition Models," in *IEEE Open Journal of the Computer Society*, vol. 3, pp. 271-280, 2022, doi: 10.1109/OJCS.2022.3218559. Url = <https://ieeexplore.ieee.org/abstract/document/9933823>.
- [5] Z. Guo, J. Xu and J. Xu, "Protecting Personal Privacy Against Unauthorized Facial Recognition Models via Image Cloaking," *2025 11th IEEE International Conference on Privacy Computing and Data Security (PCDS)*, Hakodate, Japan, 2025, pp. 57-64, doi: 10.1109/PCDS65695.2025.00017. <https://ieeexplore.ieee.org/abstract/document/11172832>
- [6] Benoit, A., Ciobotaru, M., Lambert, P., Ionescu, B. (2011). Similarity Measurement for Animation Movies. In: Lee, KT., Tsai, WH., Liao, HY.M., Chen, T., Hsieh, JW., Tseng, CC. (eds) Advances in Multimedia Modeling. MMM 2011. Lecture Notes in Computer Science, vol 6523. Springer, Berlin, Heidelberg. https://doi.org/10.1007/978-3-642-17832-0_33.
- [7] S. S. Cheung and A. Zakhor, "Efficient video similarity measurement with video signature," *Proceedings. International Conference on Image Processing*, Rochester, NY, USA, 2002, pp. I-I, doi: 10.1109/ICIP.2002.1038101. <https://ieeexplore.ieee.org/abstract/document/1038101>